Increased Prevalence of an Oxytocin Receptor Gene Variant in Functional Neurological Disorder: A Case-Control Study

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Objective: Current models of functional neurological disorder (FND) suggest a multifactorial origin of the disorder. Recent studies have identified biological vulnerability factors for FND, such as reduced amygdalar and hippocampal volumes or altered stress responses, highlighting the need to investigate the potential roles of genetic factors in this disorder.

Methods: Eighty-five patients with mixed FND symptoms and 76 healthy control (HC) individuals were genotyped for 10 single-nucleotide polymorphisms (SNPs) in seven genes associated with the brain's stress response system. For genetic variants that were found to be linked to FND, associations with structural brain alterations were investigated by using a region-of-interest approach in a subset of FND patients with complete genotyping and neuroimaging data (N=82). Regions had been previously selected on the basis of their biological involvement and being a factor in vulnerability to FND.

Results: A significant association between FND and a SNP, rs53576, in the oxytocin receptor (*OXTR*) gene was observed, and a significant association between decreased right insular volumes and rs53576 was also identified. Among female FND patients (N=60), the rs53576 SNP in *OXTR* was associated with significantly reduced bilateral amygdalar volume.

Conclusions: These preliminary results suggest that genetic factors in the oxytocinergic system and sex-specific structural changes in the insula and amygdala contribute to vulnerability to FND. Because oxytocin is a regulatory factor in stress responses, decreased activity of some FND-associated *OXTR* gene variants might affect stress responses and regulation in FND in a sex-dependent manner.

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Functional neurological disorder (FND) comprises diverse neurological symptoms, which have been attributed to underlying functional and structural brain alterations (1), with current models aiming at integrating a multifactorial origin of FND by means of a stress-diathesis model (2). To better understand FND in a multifactorial framework, researchers need to unveil what other factors could play a role in increased vulnerability to FND. The research on the genetic contribution to FND is still in its early stages. Preliminary results from studies of FND (3) have indicated that carriers of a single-nucleotide polymorphism (SNP), rs4570625, in the tryptophan hydroxylase 2 (*TPH2*) gene, encoding a protein involved in serotonin biosynthesis, had an earlier onset of FND and altered amygdalar functional connectivity associated with childhood trauma. We have partially replicated these findings (4) by identifying a significant association between *TPH2* and FND symptom severity, as well as the *TPH1* SNP rs1800532, which is associated with worse clinical outcomes among FND patients. Another recent study with a subpopulation of the same cohort (5) specifically focused on oxytocin's involvement in FND and detected an interaction of genotype and gene methylation: in the FND population, an A allele of the SNP rs53576 in the oxytocin receptor (*OXTR*) gene was associated with higher methylation levels at *OXTR*, compared with healthy control (HC) individuals. Interestingly, no interaction of *OXTR* genotype with childhood trauma has been previously identified (5).

In this report, we investigated genetic susceptibility to FND by using a set of SNPs that we selected on the basis of their association with stress pathways, neuroplasticity, and neurotransmitters previously reported to be relevant to other stress-related and mental disorders. We also tested for a genetic association between these SNPs and structural brain

alterations in FND on the basis of a priori defined regions of interest (ROIs).

METHODS

Participants

This study was conducted at the University Hospital Inselspital Bern, Switzerland, between June 2020 and February 2022. Board-certified neurologists diagnosed FND on the basis of *DSM-5* (6) criteria and positive signs (7). Study participants provided a blood sample and filled in questionnaires during a first visit and had a neuroimaging examination during a second visit. All participants completed the Beck Depression Inventory (8), the State-Trait Anxiety Inventory (9), and the Childhood Trauma Questionnaire (CTQ [10]). For the CTQ, a total CTQ score was calculated by summing the scores of the five subscales. Further details have been previously described (2, 5). All procedures were performed in compliance with institutional guidelines and were approved by the Ethics Committee of the Canton Bern (DRKS00012992). All study participants provided written informed consent.

Genotyping

DNA was extracted from whole-blood samples by using the QIAmp DNA Blood kit (Qiagen) and quantified with the Quant-it dsDNA Broad-Range Assay Kit (Invitrogen). Each sample was adjusted to 60 µL with a DNA concentration of 20 ng/µL.

The genotyping was conducted at the Spanish National Center for Genotyping. Primers were designed with MassARRAY Assay Design Software (see Table S1 in the online supplement to this report). DNA samples were genotyped with the iPLEX Assay (11), followed by mass spectrometry analysis with the MassARRAY System (Agena Bioscience). This analysis included the following steps: polymerase chain reaction amplification, with melting time adjusted for each primer; shrimp alkaline phosphatase treatment to remove unincorporated nucleotides; and single-base extension to create nucleotide mass–specific products. Processed samples were then dispensed on a 384-well SpectroCHIP bioarray and analyzed with a matrix-assisted laser desorption/ionization time-of-flight technique within the MassARRAY Workstation.

Statistical analyses were performed in R, version 4.2.1; the alpha level was set at 0.05. We selected seven genes for which 10 SNPs were analyzed, including the *BDNF* (brain-derived neurotrophic factor) gene, two dopamine receptor genes (*DRD2* and *DRD4*), FK506 binding protein 5 (*FKBP5*) gene, *TPH1* and *TPH2* genes, and the *OXTR* gene. Genotype call rate, minor allele frequency, and Hardy-Weinberg equilibrium (HWE), assessed with chi-square tests, were calculated as quality-control measures (see Table S2 in the online supplement).

We investigated associations of specific SNPs with an FND diagnosis in a between-group analysis. For each SNP, a logistic regression was performed in SNPassoc (12); age, sex, depression, and trait-anxiety and total CTQ scores were included as covariates. Three different models of inheritance were considered: codominant, dominant, and recessive. Odds ratios and confidence intervals were calculated to evaluate the statistical significance of associations between group and genotype. Because of the high prevalence of FND among females (13) and the differential impact of biological sex on genetics and endocrinology (14), analyses were repeated in females only. No correction for multiple comparisons was applied, in line with the candidate gene approach and exploratory design of this pilot study.

Brain Alterations

Anatomical images were acquired for all participants except for three FND patients and three HC individuals. We used a voxel-based morphometry approach implemented in the CAT12 toolbox (CAT12—http://www.neuro.uni-jena.de/cat) within SPM12 (https://www.fil.ion.ucl.ac.uk/spm/software/spm12). Structural T1-weighted MRI data were preprocessed by using standard CAT12 pipelines (15–18), including spatial normalization via Dartel (Diffeomorphic Anatomical Registration Through Exponentiated Lie Algebra) and tissue segmentation and modulation. All images were visually inspected for sufficient quality, and ROI volume data were screened for statistical outliers, defined as values ≥3 SDs from the group mean; no outliers were detected. Data were subsequently smoothed with an isotropic 8-mm Gaussian kernel. ROI analyses were performed with masks derived from the AAL3 atlas (19). On the basis of our previous work on oxytocin (5) and biological vulnerability factors (2), we selected the hippocampus, amygdala, and the insula as ROIs. For genetic association analyses, mean gray matter volumes were extracted from the predefined ROIs. Total intracranial volume, age, sex, depression, and trait-anxiety and total CTQ scores were included as covariates. Further acquisition and preprocessing details are provided in the online supplement.

RESULTS

Clinical Characteristics

Mixed symptom types were present among FND patients, including motor symptoms (*ICD-10* code F44.4), sensory symptoms (F44.6), functional seizures (F44.5), mixed symptom type (F44.7), and persistent postural-perceptual dizziness (see Tables S3 and S4 in the online supplement).

Genotypes and FND

Genotype data from 85 FND patients and 76 HC individuals were included in the genetic analysis (genotype distribution and sample size per group are reported in Table 1). Genotype success rate was 100% for all participants, indicating no genotyping errors, and SNPs were in HWE,

TABLE 1. Associations between *OXTR* rs53576 single-nucleotide polymorphisms and FND[a]

Model and genotype grouping	Full sample					Females only				
	FND (N=85)	HC (N=76)	OR	95% CI	p	FND (N=63)	HC (N=55)	OR	95% CI	p
Recessive										
G/G-G/A (reference)	68	70	1.00		0.012	49	50	1.00		0.022
A/A	17	6	4.17	1.31–13.23		14	5	4.21	1.17–15.11	
Codominant										
G/G (reference)	29	28	1.00		0.023	20	20	1.00		0.044
G/A	39	42	0.62	0.26–1.47		29	30	0.57	0.19–1.75	
A/A	17	6	3.16	0.90–11.10		14	5	3.09	0.75–12.75	

[a] Results of linear regression; statistical significance was set at p<0.05. FND, functional neurological disorder; HC, healthy control individuals; *OXTR*, oxytocin receptor gene.

indicating that the observed frequencies in the sample did not differ from the expected frequencies in a large, stable population (p>0.05; Table S2 in the online supplement). Using logistic regression, we identified a significant association between FND and the rs53576 SNP in the *OXTR* gene. In the recessive model, the odds of this association was OR=4.17 (p=0.012). In the codominant model, the odds of an association with FND was 3.16 for the A/A genotype and 0.62 for the G/A genotype (p=0.023). Analyses including only female patients gave similar results (Table 1). None of the other SNPs showed a significant association with FND.

Genotypes and Brain Alterations

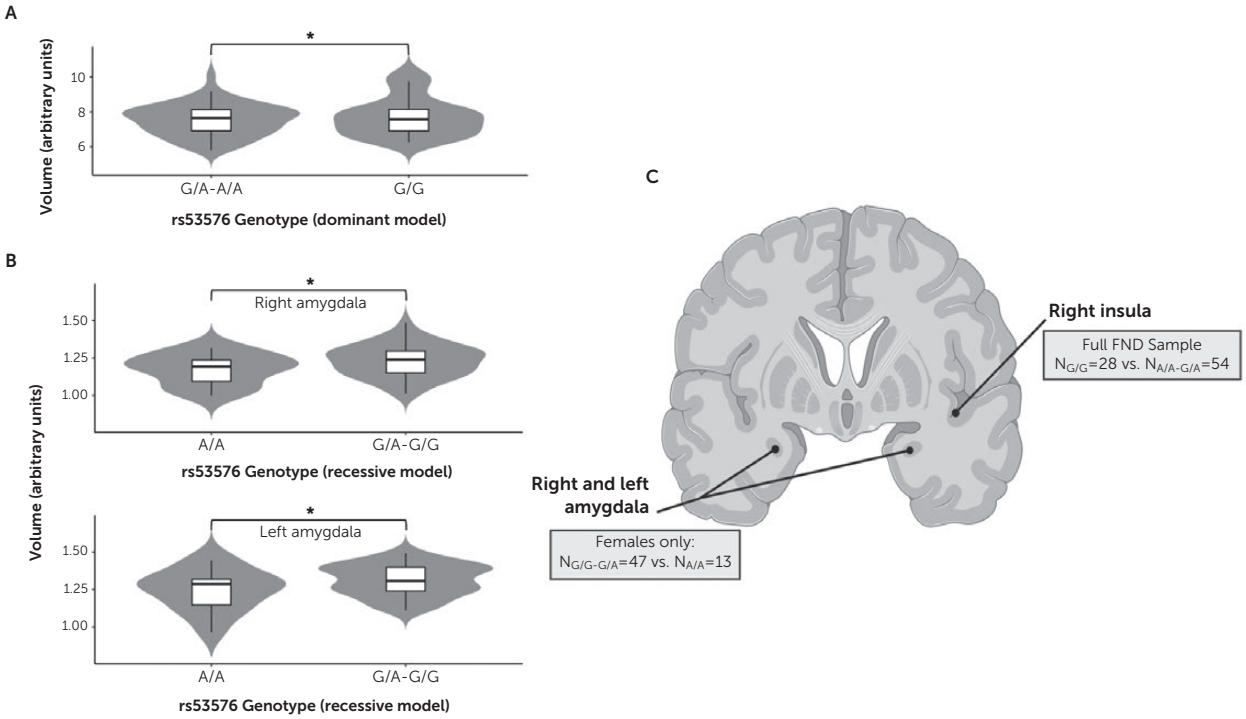
The final sample for neuroimaging analyses included 82 FND patients and 73 HC individuals with complete structural MRI and genotyping data (genotype distribution and sample size per group are reported in Table S2 in the online supplement). For FND patients, a significant association was found between the right insular volume and the rs53576 SNP of the *OXTR* gene (dominant model: β=−0.37, SE=0.1, p=0.04), with carriers of the minor rs53576 A allele having smaller insular volumes. A significant association was identified for female FND patients, but not for the entire cohort or HC individuals, between the bilateral amygdala and the rs53576 SNP in the *OXTR* gene (left amygdala, recessive model: β=−0.05, SE=0.04, p=0.02; right amygdala, recessive model: β=−0.04, SE=0.03, p=0.046), indicating that female patients with FND who were homozygous carriers (A/A) of the *OXTR* rs53576 A allele had smaller volumes in the bilateral amygdala, compared with female FND patients who were either heterozygous (A/G) or homozygous (G/G) carriers of the G allele. No other significant associations were detected between *OXTR* genotypes and brain ROIs (Figure 1 and Table S5 in the online supplement).

DISCUSSION

We identified a significant association between the minor A allele of the rs53576 *OXTR* SNP and FND, indicating that carriers of this gene variant might be at increased risk for FND. This variant has been reported to be associated with stress predisposition, with A-allele carriers having an increased sensitivity to stress, interacting with contextual factors like biological sex, psychological resources, or attachment style (20–22). The oxytocinergic system is suggested to regulate the stress response in the neuroendocrine and autonomic nervous systems, reducing activity in the hypothalamic-pituitary-adrenal axis (23). Because oxytocin may buffer against the effects of stress, a decrease in the expression or activity of the oxytocin receptors—due to *OXTR* SNPs—might increase the impact of stress on brain function and structure and result in stress-related psychopathology. We note that many of the associations of the rs53576 A allele with increased risk for psychopathology such as increased fear and anxiety have been reported in studies that did not always consider various environmental factors as potential moderators of this association (24).

It has been proposed that rs53576 *OXTR* SNPs have a role in environmental adaptation via genotype-specific modulation of striatal dopamine transporter availability and its interaction with oxytocin levels (25). Thus, rs53576 *OXTR* SNPs might have modulatory effects on oxytocin's contributions to stress-related psychopathology and social behavior. Results from previous studies (21, 26, 27) indicate that, particularly among *OXTR* rs53576 A-allele carriers, threat exposure in early life could contribute to general psychopathology and psychiatric disorders. It has also been reported that this gene-environment interaction could be mediated by *OXTR* methylation in a sex-specific manner (5, 28). Therefore, our results appear to align with the hypothesis that enhanced *OXTR* methylation, potentially influenced by *OXTR* genotype, could increase sensitivity to the impact of childhood trauma among female individuals (5, 28, 29). We found that the minor A allele of the *OXTR* rs53576 SNP was associated with reduced right insular brain volumes, which might be a risk factor for FND. We are not aware of studies that have associated specific *OXTR* variants with insular volume, but many studies have reported that oxytocin affects insular activity. Oxytocin related to insular activity seems to be involved in strengthening bottom-up detection of salient signals in a sex-dependent manner (30).

FIGURE 1. Structural brain differences associated with *OXTR* rs53576 genotypes among patients with FND[a]



[a] A: normalized gray matter volume in the right insula in the full functional neurological disorder (FND) sample (N=82), grouped by oxytocin receptor (*OXTR*) rs53576 genotype by using a dominant genetic model (G/A-A/A vs. G/G; $N_{G/A-A/A}$=54, $N_{G/G}$=28). Shaded areas in the graph indicate the distribution density of brain volume values, illustrating the spread and central tendency of the data. B: normalized gray matter volumes in the bilateral amygdala of female FND patients only (N=60), grouped by rs53576 genotype by using a recessive model (G/A-G/G vs. A/A; $N_{G/A-G/G}$=44, $N_{A/A}$=16). Shaded areas in the graph indicate the distribution density of brain volume values. C: brain regions showing *OXTR* rs53576 genotype–related volume differences among FND patients. Volume in the right insula was significantly reduced in A-allele carriers, compared with homozygous (G/G) carriers of the G allele, in the dominant genetic model. For female FND patients, bilateral amygdala volumes were significantly reduced among those homozygous for the A allele, compared with female patients who were either heterozygous (A/G) or homozygous (G/G) carriers of the G allele. Only FND patients are shown because of an absence of significant associations for healthy control individuals. (Detailed results are provided in Table S5 in the online supplement.)

Consistent with these observations, the *OXTR* rs53576 A/A genotype was associated with smaller volumes of the bilateral amygdala in female FND patients. Because oxytocin and its receptor in the amygdala have an important role in fear and anxiety (31), the observed alterations in amygdalar volume might be relevant for the modulation of neural dynamics that depend on oxytocin. The genotype-dependent and sex-specific effects on amygdalar volume should be investigated, particularly given the predominance of female patients in FND populations (13) and previous findings suggesting that the *OXTR* rs53576 variant may sex-specifically protect against stressful experiences (21). Stress has been shown to influence amygdalar structure depending on developmental trajectory and sex (32), which is crucial to consider when evaluating the distinct amygdalar alterations observed in FND populations (33). A study with female patients with functional or dissociative seizures (34) revealed a lower volume in the right lateral amygdala, along with higher right central and medial and left anterior amygdalar volumes in these patients. These findings underline the importance of considering sex as a factor that could significantly interact with the genetic vulnerability for stress-related disorders such as FND.

Although we did not test for interactions of *OXTR* rs53576 SNPs and methylation rates with childhood trauma in the present study, results of a previous study with the same cohort (5) indicated no significant association of the oxytocinergic system with childhood trauma. However, we note that previous findings have indicated some variability in this association, including that the rs53576 genotype or its interaction with emotional neglect has no effects on amygdalar and hippocampal volumes (35), that gray matter volume of the left amygdala is jointly modulated by *OXTR* rs53576 genotype and childhood attachment security (36), and that trauma-exposed youths have increased salience network connectivity within the left amygdala (37). The authors of another study (38) in which interactions between emotional trauma and *OXTR* rs53576 had no effects on bilateral amygdala volume suggest that this SNP may have a differential impact on brain volume, depending on age, sex, and type of environmental exposure.

Even though our results link the *OXTR* rs53576 minor A allele to FND, we note some discrepancies in previous studies regarding the roles of the G allele as a risk allele in stress and mental disorders (39). A-allele carriers have been shown

to have a higher risk for mental illness and higher sensitivity to stress, which in some cases also involves interactions with an individual's ability for emotional regulation and exposure to childhood trauma (21, 40–42). Moreover, the G allele has been associated with an increased risk for antisocial behavior among adolescent females exposed to maltreatment (43) or with lower perceived social support and higher levels of internalizing symptoms among maltreated adolescents (44). These observations also hint at the complex interactions of genetic, environmental, and developmental factors, with factors such as stress sensitivity and reactivity, environmental risk factors (39), and neuronal alterations possibly playing a role in FND (45). Several factors should be considered when interpreting our results, such as sex-specific interactions, childhood trauma, age, and differences in *OXTR* rs53576 genotype distribution among individuals from different ethnicities (46). Moreover, other genes and SNPs not assessed in this study may be involved in FND severity and sex-specific manifestation.

A major limitation of our study was its small sample size, a common challenge in genetic studies to identify significant clinical-genetic associations. Furthermore, using a candidate gene approach and ROI analyses may have limited our ability to detect broader genetic effects or interactions and may have increased the risk for false-positive findings. Because of the exploratory and hypothesis-driven nature of this pilot study, we applied no formal correction for multiple comparisons to the genetic or neuroimaging analyses. Instead, we focused on a small set of SNPs and ROIs selected a priori on the basis of existing literature implicating oxytocinergic signaling and limbic structures in the pathophysiology of FND. Another relevant limitation was potentially confounding effects of lifetime comorbid psychiatric conditions. Although we controlled for current depressive symptoms in our analyses, we could not exclude the influence of other current or past psychiatric diagnoses, which could affect the interpretation of genotype-phenotype associations. We did not attempt to comprehensively investigate genotype-environmental interactions; we therefore did not directly examine interactions of gene SNPs with risk factors such as childhood trauma, which we previously analyzed in the same cohort (5). Nevertheless, the candidate gene approach we used was appropriate given the sample size and the exploratory nature of this study, in which we chose to investigate specific brain regions previously shown to be involved in the pathophysiology of FND in order to explore the susceptibility for FND at the genetic level.

CONCLUSIONS

Given that oxytocin is part of a key stress response system in the brain, decreased sensitivity of some *OXTR* genetic variants could have a significant impact on stress regulation in FND, particularly for women. Moreover, reduced volumes in the right insula of FND patients might represent vulnerability factors, with a potential genetic component dependent on sex.

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